

3D Spectral Imaging for LIBS Elemental Analysis in Heritage

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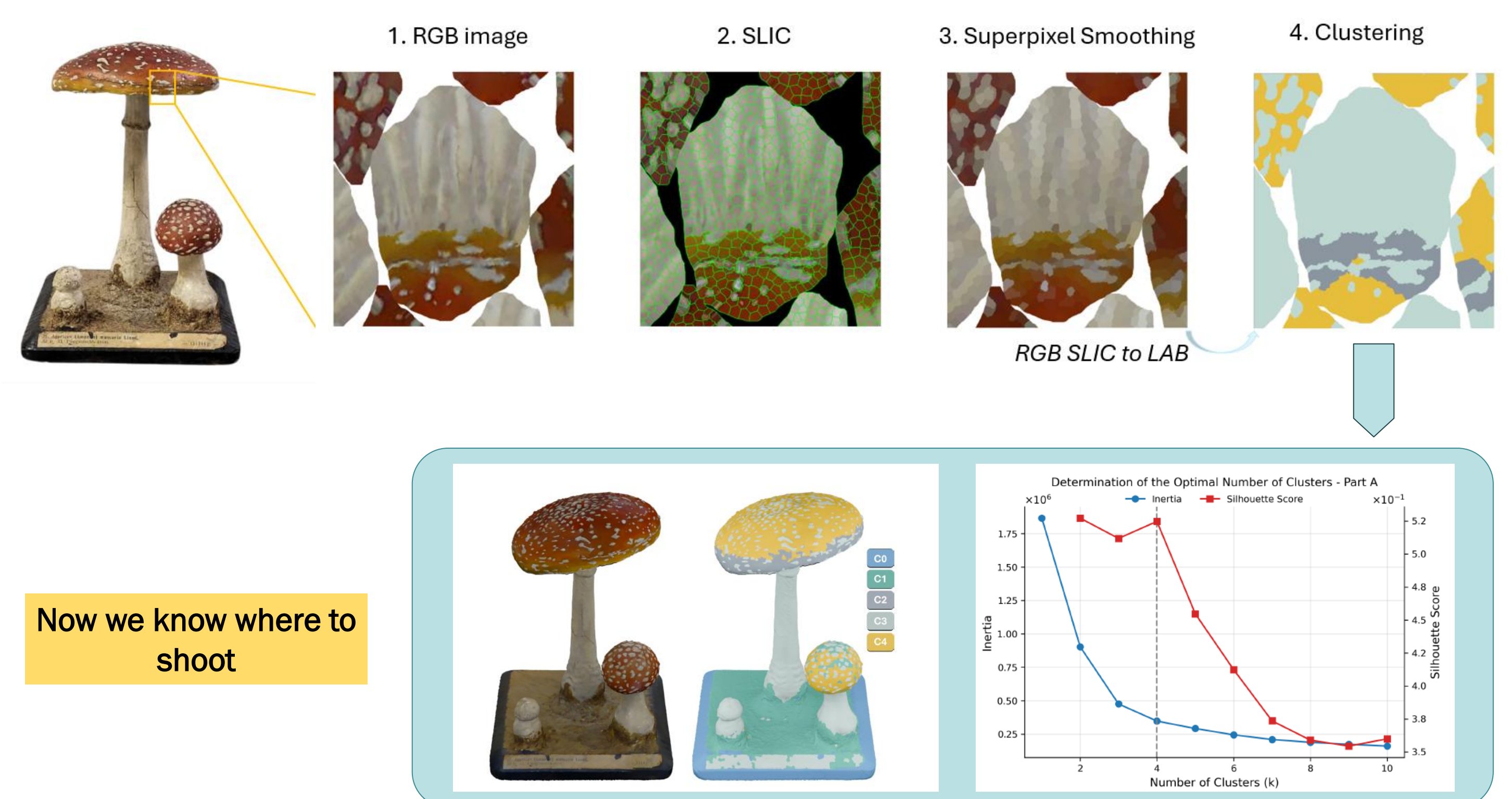
Motivation The integration of spatial geometry and chemical composition remains a key challenge in the analysis of heterogeneous and fragile materials, particularly in cultural heritage, where minimally invasive approaches are essential. In this work, a 3D spectral imaging methodology combining Laser-Induced Breakdown Spectroscopy (LIBS) with 3D scanning is proposed, enabling spatially resolved elemental analysis while minimizing sample damage.

Key Ideas

A clustering strategy is first applied to RGB images, segmenting the surface into regions of similar colour, which are assumed to correspond to chemically homogeneous areas. This clustering methodology was previously validated for quasi-2D ceramic tiles in our earlier work [1]. In the present study, the approach is extended to complex 3D objects by first acquiring the surface texture with a Creality RaptorX 3D scanner and mapping it onto the reconstructed geometry. This enables a direct correspondence between the 2D image coordinates, where clustering is performed, and the 3D model.

Spectral information was acquired using LIBS. Instead of performing full-resolution LIBS mapping, spectra are acquired from a reduced number of representative points per cluster, significantly decreasing the number of laser shots. Elemental distributions are then reconstructed by assigning cluster-wise spectral information to the segmented regions.

Clustering Methodology



Instrumentation

Figure 1. LIBS setup and parameters

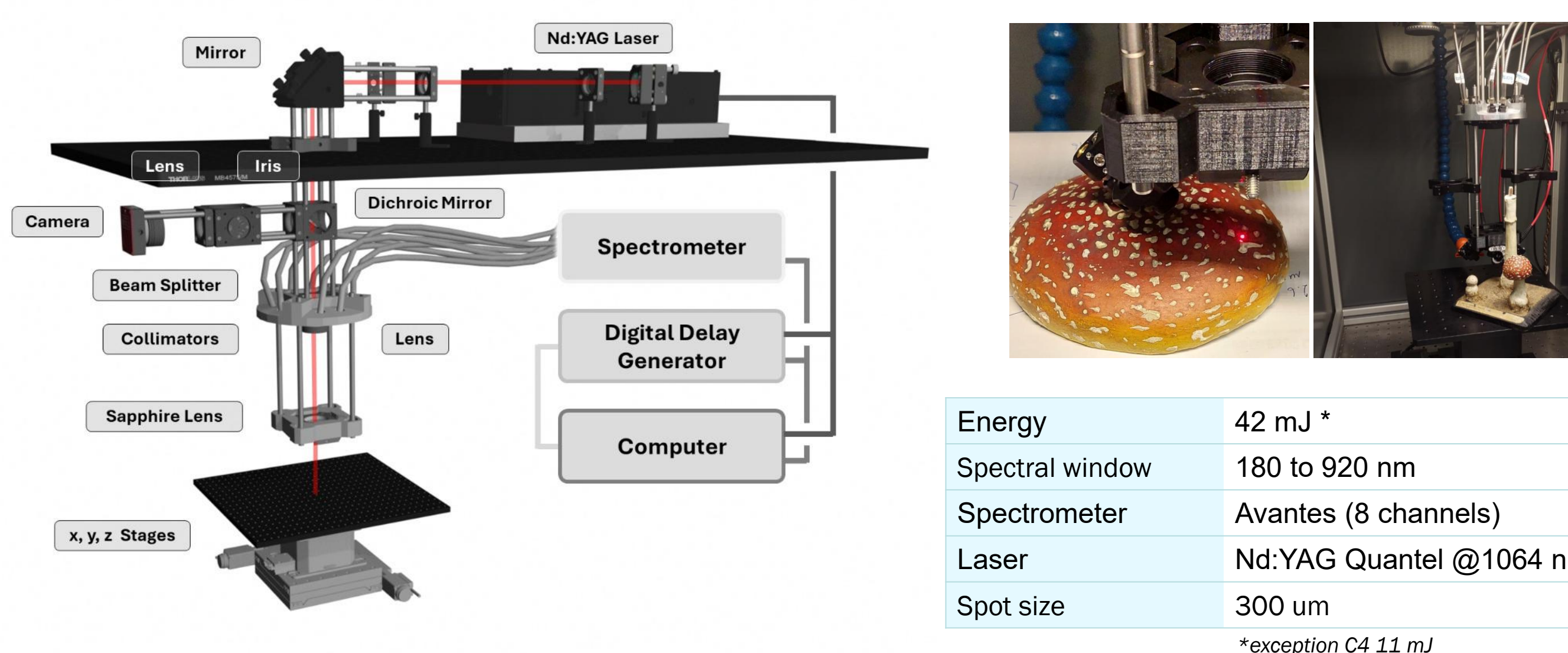


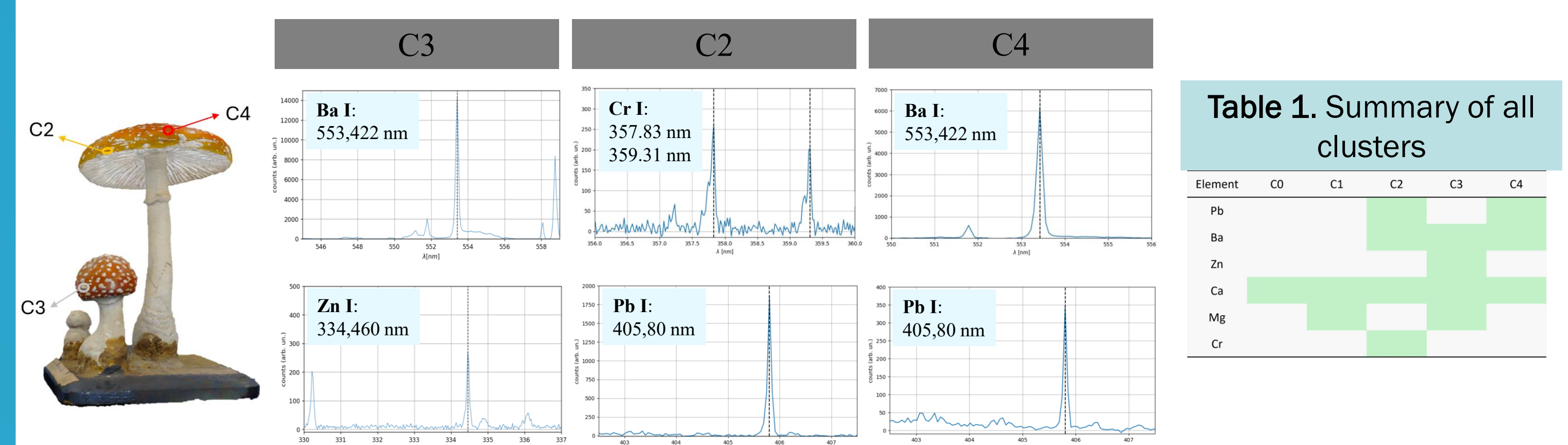
Figure 2. Creality 3D Scanner



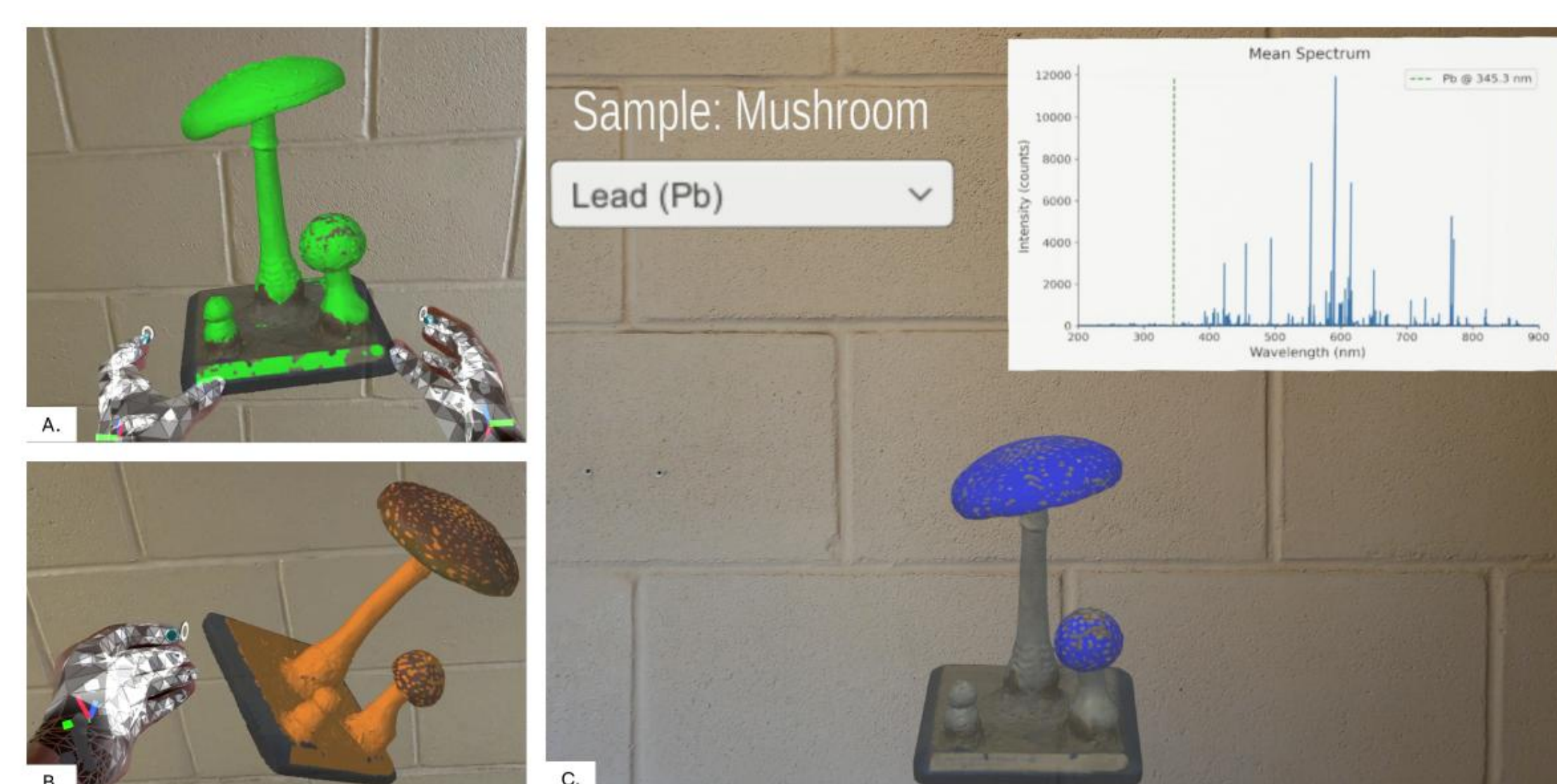
Figure 3. Hololens 2 Augmented Reality glasses



Preliminary LIBS results



Visualization – Augmented Reality



Sample interaction: move, shrink, stretch, rotate a sample

Canva interaction: exploring information contained in the mean spectrum and change the cluster represented over the sample

Menu interaction: fully interactive environment and ability to change samples

Sample Description

The University of Porto Herbarium houses a unique collection of 45 papier-mâché mushroom models from the renowned manufacturer H. Arnoldi (19th century). These life-size didactic models were used for over a century to teach students about the diversity of fungi and distinctions between edible and poisonous species. This collection, the only one of its kind in Portugal, represents a valuable piece of scientific and educational history.

Figure 4. *Boletus edulis* model *Amanita Muscaria*



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Conclusions

By integrating elemental information with the geometry obtained from surface scanning, the proposed framework enables the generation of chemically resolved digital twins, allowing intuitive visualization and interpretation of both morphology and chemical composition. More importantly, this work unifies two complementary contributions into a single workflow: (i) RGB-guided clustering for representative sparse LIBS sampling, minimizing analytical damage, and (ii) geometry-aware integration of spectroscopic data through 3D reconstruction and augmented reality. This unified workflow provides a practical foundation for non-destructive, geometry-aware spectroscopic analysis of complex three-dimensional objects, supporting advanced 3D chemical imaging for cultural heritage applications.

[1] R. Cavaco, D. Capela, P. A. S. Jorge, N. A. Silva, D. Guimarães, Minimizing LIBS damage in the analysis of decorative tiles using RGB data clustering. Journal of Cultural Heritage, 78 (2026), 175-186. <https://doi.org/10.1016/j.culher.2026.02.001>

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